AI Learning Navigator – Student Performance Analysis System

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1. OBJECTIVE:

The objective of this project is to develop an AI-driven system that personalizes learning for students based on their academic performance, attendance, and preparation levels.

The system aims to move beyond traditional one-size-fits-all education by analyzing student data and generating personalized learning paths, recommendations, and career alignment insights.

2. PROBLEM STATEMENT:

Educational institutions manage diverse student populations with varying learning speeds, strengths, and career aspirations.

Traditional education systems often follow standardized curricula, which:

- Fail to adapt to individual learning needs
- Overlook different cognitive and learning styles
- Do not align strongly with evolving job market demands

Educators and academic planners require deeper insights into:

• Identifying students who are at risk of falling behind or disengaging
• Adapting learning pace and content difficulty for individual students
• Recognizing skills relevant for future career success
• Personalizing learning journeys while maintaining curriculum standards

The goal of this project is to build an AI-driven system that dynamically generates personalized learning paths, improves student performance, and enhances future career readiness.

3. METHODOLOGY:

The project follows a structured data-driven approach:

1. Data Preprocessing:

- Handled missing values in test preparation
- Standardized categorical data

2. Exploratory Data Analysis (EDA):

- Study time vs performance analysis
- Absence impact analysis
- Test preparation comparison
- Correlation heatmap

3. **Clustering:**

- Applied KMeans clustering
- Segmented students into:
  - Low Performer
  - Average (At Risk)
  - High Performer

4. **Recommendation System:**

- Rule-based logic using multiple conditions
- Generated personalized suggestions

5. **Skill Gap Analysis:**

- Identified student learning levels:
  - Weak Fundamentals
  - Needs Improvement
  - Strong

6. **Learning Path Generation:**

- Personalized paths based on performance level

7. **Career Alignment:**

- Suggested career directions based on performance and consistency

8. **Action Plan:**

- Assigned priority levels:
  - High Priority
  - Medium Priority
  - Low Priority

4. FLOW DIAGRAM:

5. SAMPLE INPUT:

math_score, absences, study_time_weekly, test_preparation

6. SAMPLE OUTPUT:

math score	performance_ level	skill_ gap	learning_path	career_ path	action_ plan
99	High Performer	Strong	Advanced learning	Data Science	Low Priority
86	Average (At Risk)	Strong	Improve consistency	Needs consistency	Medium Priority
83	High Performer	Strong	Skill Development	Focus on Foundation	Low Priority

7. Dashboard Implementation:

An interactive dashboard was developed using Streamlit to visualize and analyze student performance data in real time. The dashboard acts as the front-end interface of the AI Learning Navigator system, integrating machine learning outputs, data visualization, and AI-based recommendations into a unified platform.

The system processes student data, applies clustering techniques for segmentation, and dynamically updates visualizations based on user-defined filters. This allows users to explore patterns and insights efficiently.

Key Features:

• Interactive sidebar filters for:

- Performance level
- Score range
- Absence range

• Real-time updates across all visual components based on selected filters

• Student segmentation using K-Means clustering:

- Low Performer
- Average (At Risk)
- High Performer

• AI-powered recommendation system:

- Cohort-level recommendations based on average performance
- Student-level recommendations based on individual behavior

• Data visualizations including:

- Pie chart for performance distribution
- Bar chart for category-wise student count
- Scatter plots:
    - Study time vs performance
    - Absences vs performance
- Skill gap distribution chart

• Identification of:

- Top-performing students
- At-risk students

• Tab-based interface for structured navigation:

- Home (Overview + AI insights)
- Analysis (Behavioral relationships)
- Insights (Skill gap distribution)
- Students (Detailed dataset view)

Purpose:

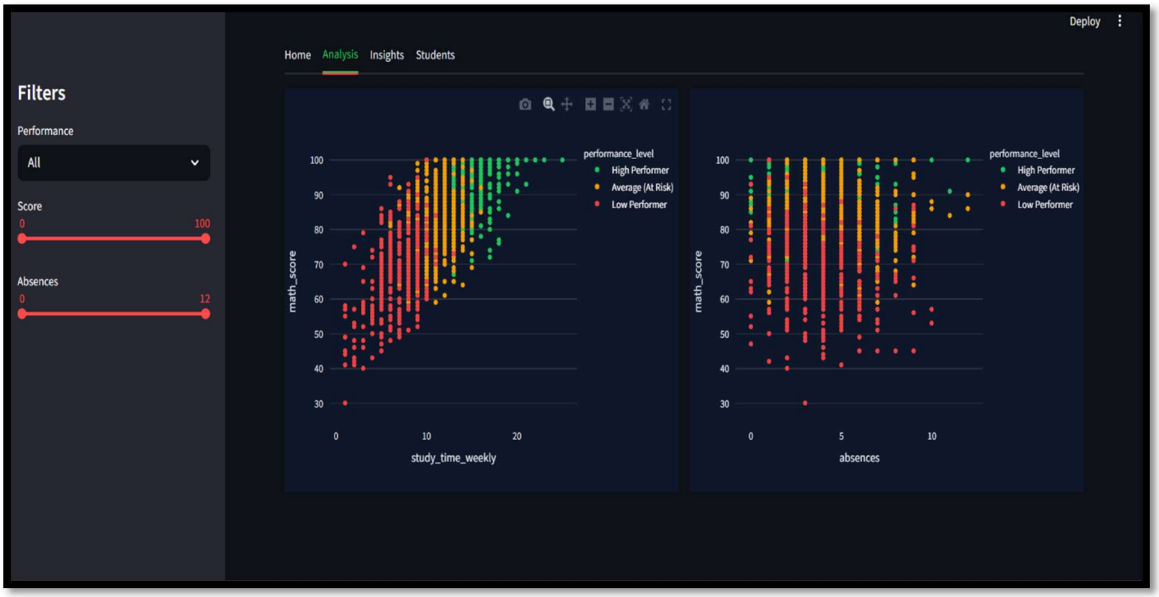
The dashboard enhances decision-making by converting static academic data into an interactive and intelligent system.

It enables users to:

• Monitor student performance in real time
• Identify critical student groups such as “Average (At Risk)”
• Analyze the impact of study habits and attendance on performance
• Detect skill gaps across different performance levels
• Generate actionable recommendations using AI logic

By combining clustering, rule-based AI recommendations, and interactive visualization, the system demonstrates a practical implementation of AI-driven personalized learning. It provides a scalable solution for educational institutions to improve student outcomes and optimize learning strategies.

Screenshots:


Deploy
Home
Analysis
Insights
Students
Filters
Performance
All
Score
0
100
Absences
0
12
performance_level
High Performer
Average (At Risk)
Low Performer
math_score
study_time_weekly
absences

Deploy
Home
Analysis
Insights
Students
Filters
Performance
All
Score
0
100
Absences
0
12
skill_gap
Strong
Needs Improvement
Weak Fundamentals
count
Skill: Needs Improvement
Count: 372

Deploy

Home Analysis Insights Students

	student_id	course_id	age	gender	ethnicity	study_time_weekly	absences	test_preparation	math_score	reading_score	writing_score	science_score
0	1	C104	20	Male	Group C	13	2	Completed	99	96	83	86
1	2	C105	18	Male	Group B	13	12	Completed	86	100	68	85
2	3	C103	20	Male	Group A	9	2	None	83	82	87	67
3	4	C105	19	Male	Group C	15	2	Completed	96	100	100	100
4	5	C105	17	Male	Group B	12	2	None	89	77	83	81
5	6	C102	20	Female	Group B	16	9	Completed	100	94	76	90
6	7	C103	17	Female	Group B	17	1	Completed	89	93	99	91
7	8	C103	18	Female	Group C	6	4	None	74	73	89	68
8	9	C103	17	Female	Group D	8	7	None	71	83	76	63
9	10	C105	18	Female	Group C	15	3	Completed	85	95	100	88

Top Performers

	student_id	course_id	age	gender	ethnicity	study_time_weekly	absences	test_preparation	math_score	reading_score	writing_score	science_score
5	6	C102	20	Female	Group B	16	9	Completed	100	94	76	90
384	385	C105	17	Female	Group A	21	3	None	100	100	100	100
385	386	C101	17	Female	Group D	12	3	Completed	100	86	97	80
330	331	C103	19	Male	Group E	10	1	Completed	100	83	74	85
331	332	C102	18	Female	Group E	15	2	Completed	100	87	83	77
389	390	C102	18	Male	Group D	17	6	Completed	100	91	100	90
391	392	C105	19	Female	Group C	20	4	None	100	100	100	100
394	395	C105	17	Male	Group E	14	1	Completed	100	88	98	98
396	397	C105	21	Male	Group B	19	7	Completed	100	95	100	96
403	404	C105	17	Female	Group C	13	7	None	100	100	100	100

8. KEY INSIGHTS:

1. Student performance is multi-dimensional and depends on study habits, attendance, and preparation.

2. Study time has a strong positive impact on performance.

3. Absences alone are not a strong predictor but affect consistency.

4. The “Average (At Risk)” group is the most critical segment, as these students have high potential but inconsistent behavior.

4. Consistency plays a more important role than raw performance scores in long-term success.

9. AI-BASED INTERVENTION STRATEGY:

The system integrates clustering, recommendation logic, skill gap analysis, and career alignment to form a complete decision-making framework.

Instead of just analyzing data, it provides actionable insights and personalized guidance for each student, making it suitable for real-world educational applications.

10. CONCLUSION:

This project demonstrates that student performance cannot be evaluated using a single metric. By combining multiple factors such as study habits, attendance, and preparation, the system provides a comprehensive understanding of student behaviour.

The integration of clustering, personalized recommendations, skill gap analysis, and career alignment transforms raw data into meaningful insights.

The identification of the “Average (At Risk)” group highlights a key opportunity for targeted intervention, making the system practical, scalable, and impactful for modern education systems.